BI DIRECTIONAL SHIFT REGISTER

UNIVERSAL SHIFT REGISTER

RING COUNTER

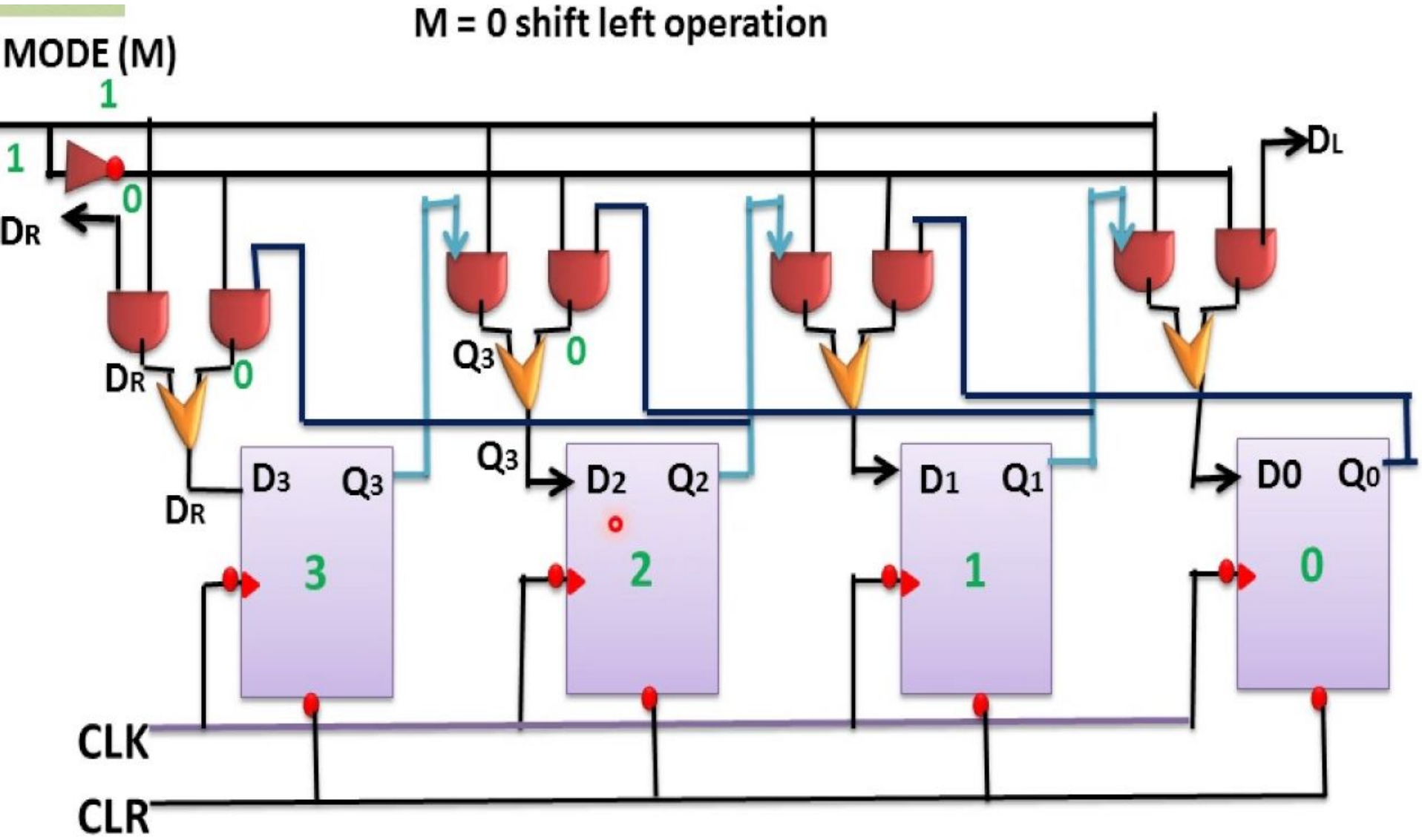
JOHNSON'S COUNTER

# BI DIRECTIONAL SHIFT REGISTER

Bidirectional shift registers are the registers which are capable of shifting the data either right or left depending on the mode selected. If the mode selected is 1(high), the data will be shifted towards the right direction and if the mode selected is 0(low), the data will be shifted towards the left direction.

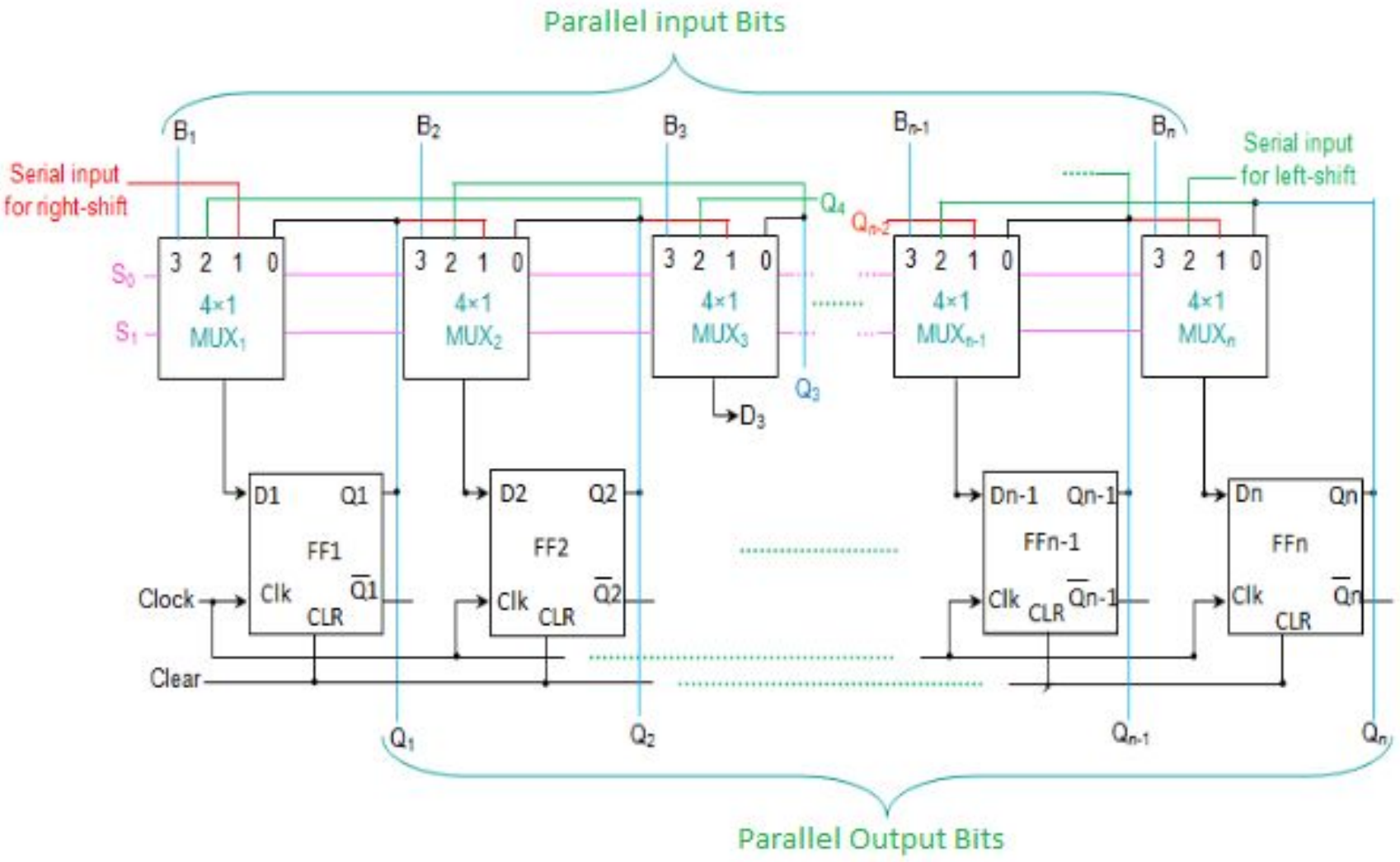
The circuit consists of four D flip-flops which are connected. The input data is connected at two ends of the circuit and depending on the mode selected only one and gate is in the active state.

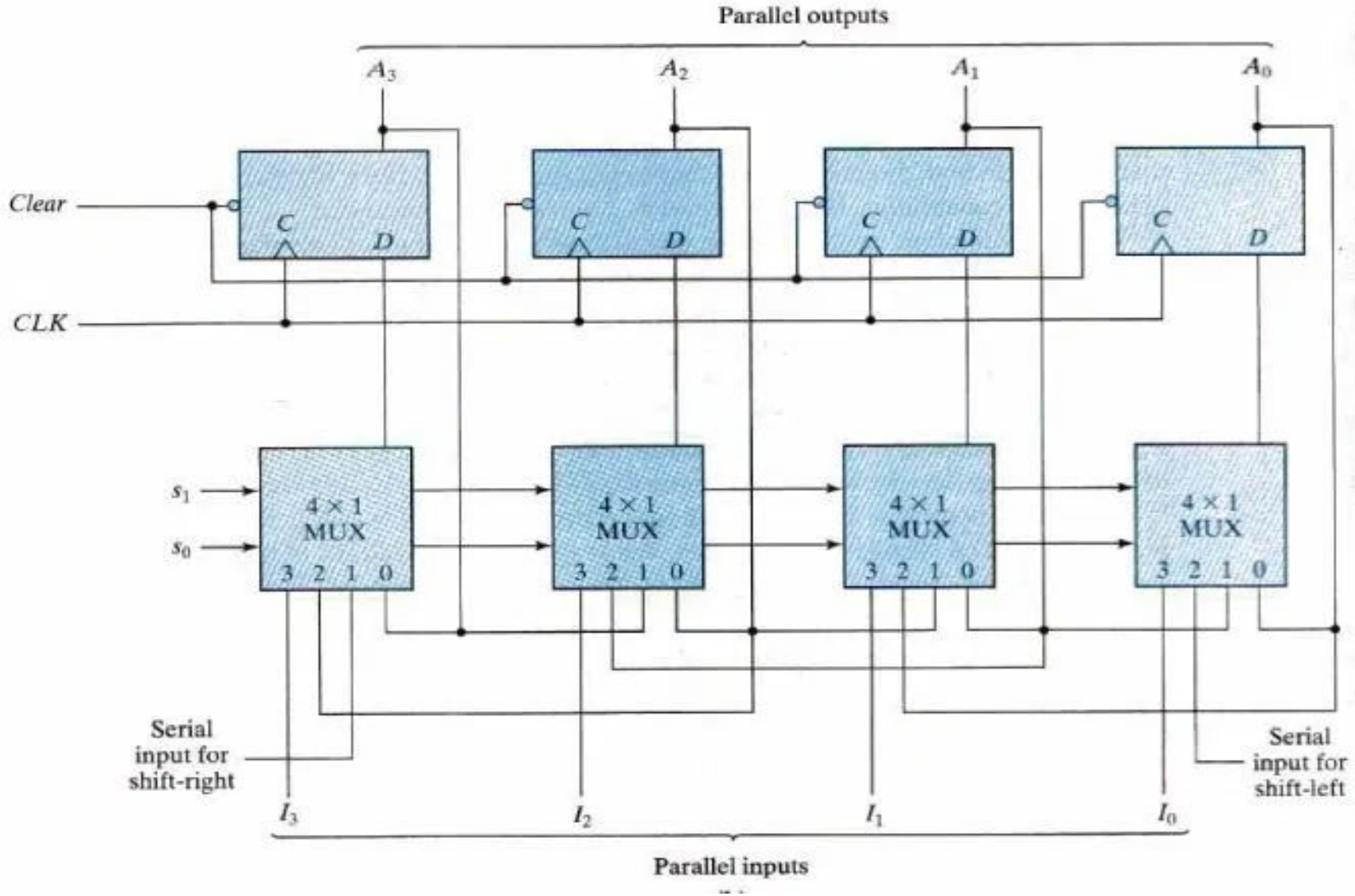
# BI DIRECTIONAL SHIFT REGISTER



If we shift a binary number to the left by one position, it is equivalent to multiplying the number by 2 and if we shift a binary number to the right by one position, it is equivalent to dividing the number by 2.To perform these operations we need a register which can shift the data in either direction.

# UNIVERSAL SHIFT REGISTER




Parallel outputs
$A_3$
$A_2$
$A_1$
$A_0$
Clear
CLK
C
D
$s_1$
$s_0$
4 × 1
MUX
3 2 1 0
Serial
input for
shift-right
Serial
input for
shift-left
$I_3$
$I_2$
$I_1$
$I_0$
Parallel inputs

A Universal shift register is a register which has both the right shift and left shift with parallel load capabilities. Universal shift registers are used as memory elements in computers. A Unidirectional shift register is capable of shifting in only one direction. A bidirectional shift register is capable of shifting in both the directions. The Universal shift register is a combination design of **bidirectional** shift register and a **unidirectional** shift register with parallel load provision.

**n-bit universal shift register –**

A n-bit universal shift register consists of n flip-flops and n 4×1 multiplexers. All the n multiplexers share the same select lines(S1 and S0)to select the mode in which the shift register operates. The select inputs select the suitable input for the flip-flops.

# Basic connections

The first input (zeroth pin of multiplexer) is connected to the output pin of the corresponding flip-flop.

The second input (first pin of multiplexer) is connected to the output of the very-previous flip flop which facilitates the right shift.

The third input (second pin of multiplexer) is connected to the output of the very-next flip-flop which facilitates the left shift.

The fourth input (third pin of multiplexer) is connected to the individual bits of the input data which facilitates parallel loading.

| S1 | S0 | Register operation |
|---|---|---|
| 0 | 0 | No changes |
| 0 | 1 | Shift right |
| 1 | 0 | Shift left |
| 1 | 1 | Parallel load |

# APPLICATIONS OF SHIFT REGISTERS

- The shift registers are used for temporary data storage.
  The shift registers are also used for data transfer and data manipulation.
- The serial-in serial-out and parallel-in parallel-out shift registers are used to produce time delay to digital circuits.
- The serial-in parallel-out shift register is used to convert serial data into parallel data thus they are used in communication lines where demultiplexing of a data line into several parallel line is required.
- A Parallel in Serial out shift register us used to convert parallel data to serial data.
- Time delays
- Ring counters

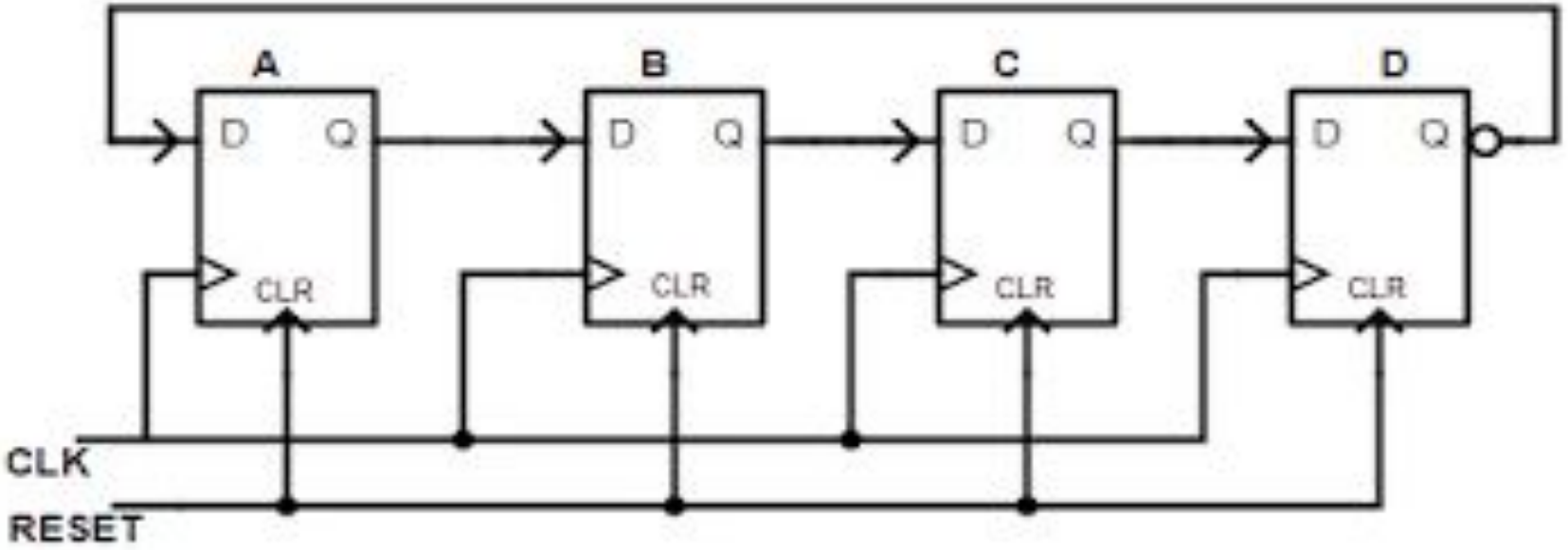
# RING COUNTER

## Shift Register Counter

Shift Register Counters are the shift registers in which the outputs are connected back to the inputs in order to produce particular sequences. These are basically of two types.

**Ring Counter –**A ring counter is basically a shift register counter in which the output of the first flip flop is connected to the next flip flop and so on and the output of the last flip flop is again fed back to the input of the first flip flop, thus the name ring counter. The data pattern within the shift register will circulate as long as clock pulses are applied.

The circuit consists of four D flip-flops which are connected. Since the circuit consists of four flip flops the data pattern will repeat after every four clock pulses.

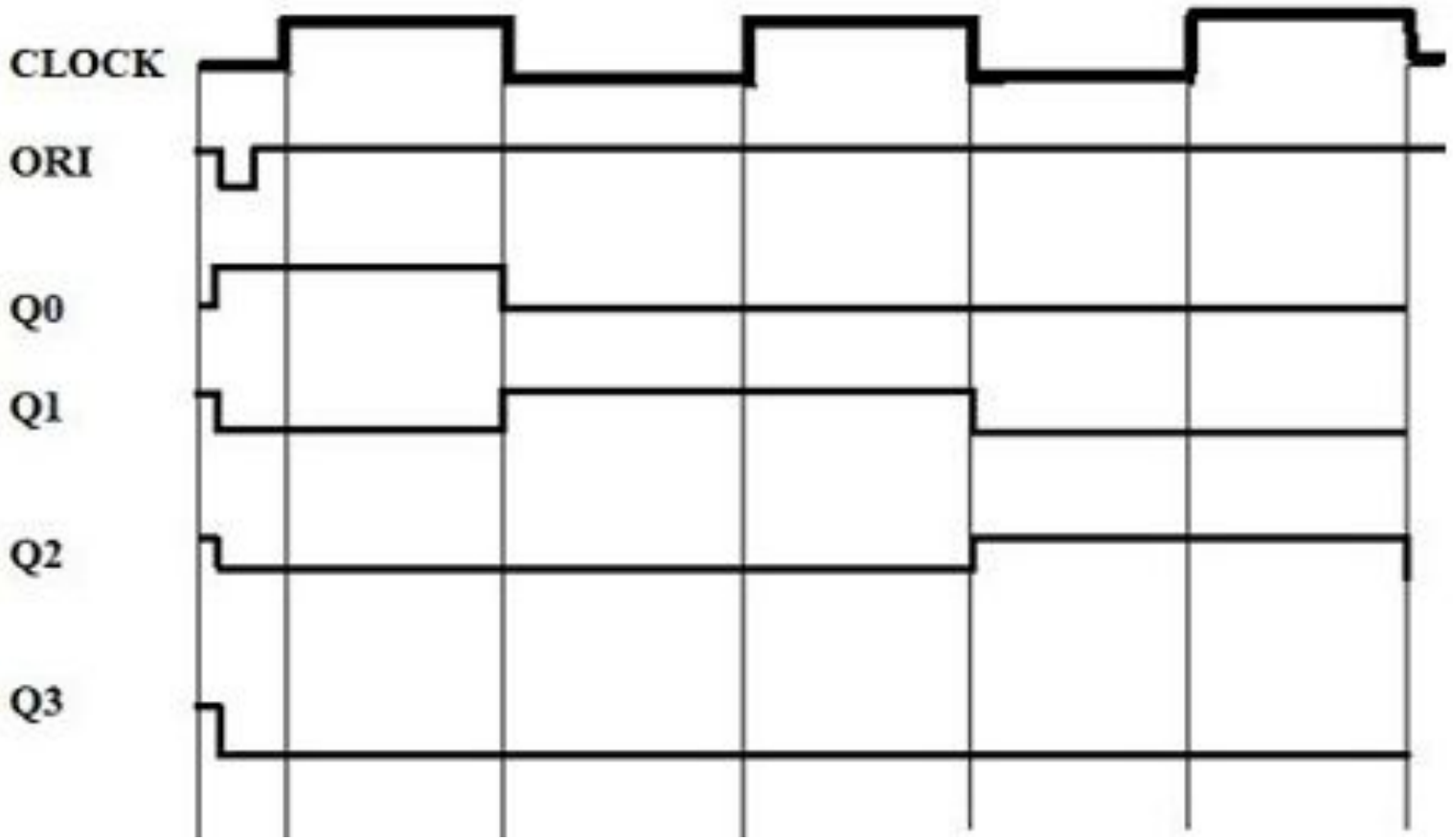



| Clock Cycle | $Q_1$ | $Q_2$ | $Q_3$ | $Q_4$ |
|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 0 |
| 2 | 0 | 1 | 0 | 0 |
| 3 | 0 | 0 | 1 | 0 |
| 4 | 0 | 0 | 0 | 1 |
| 5 | 1 | 0 | 0 | 0 |
| 6 | 0 | 1 | 0 | 0 |
| . | . | . | . | . |
| . | . | . | . | . |
| . | . | . | . | . |

Bit-pattern repeats for every 4 clock cycles

A Ring counter is generally used because it is self-decoding. No extra decoding circuit is needed to determine what state the counter is in.

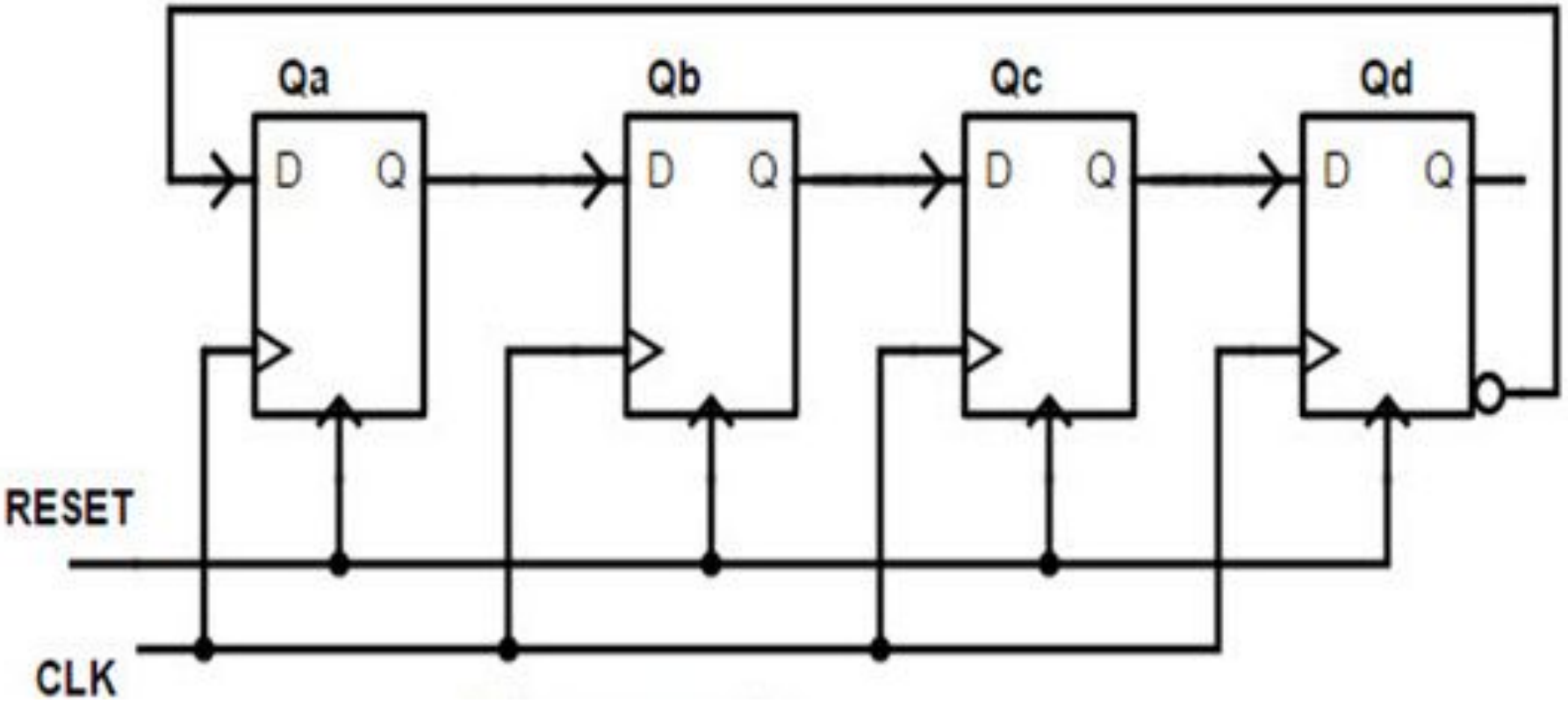
# RING COUNTER TIMING DIAGRAM



# JOHNSON's COUNTER(Twisted / switch Tail)

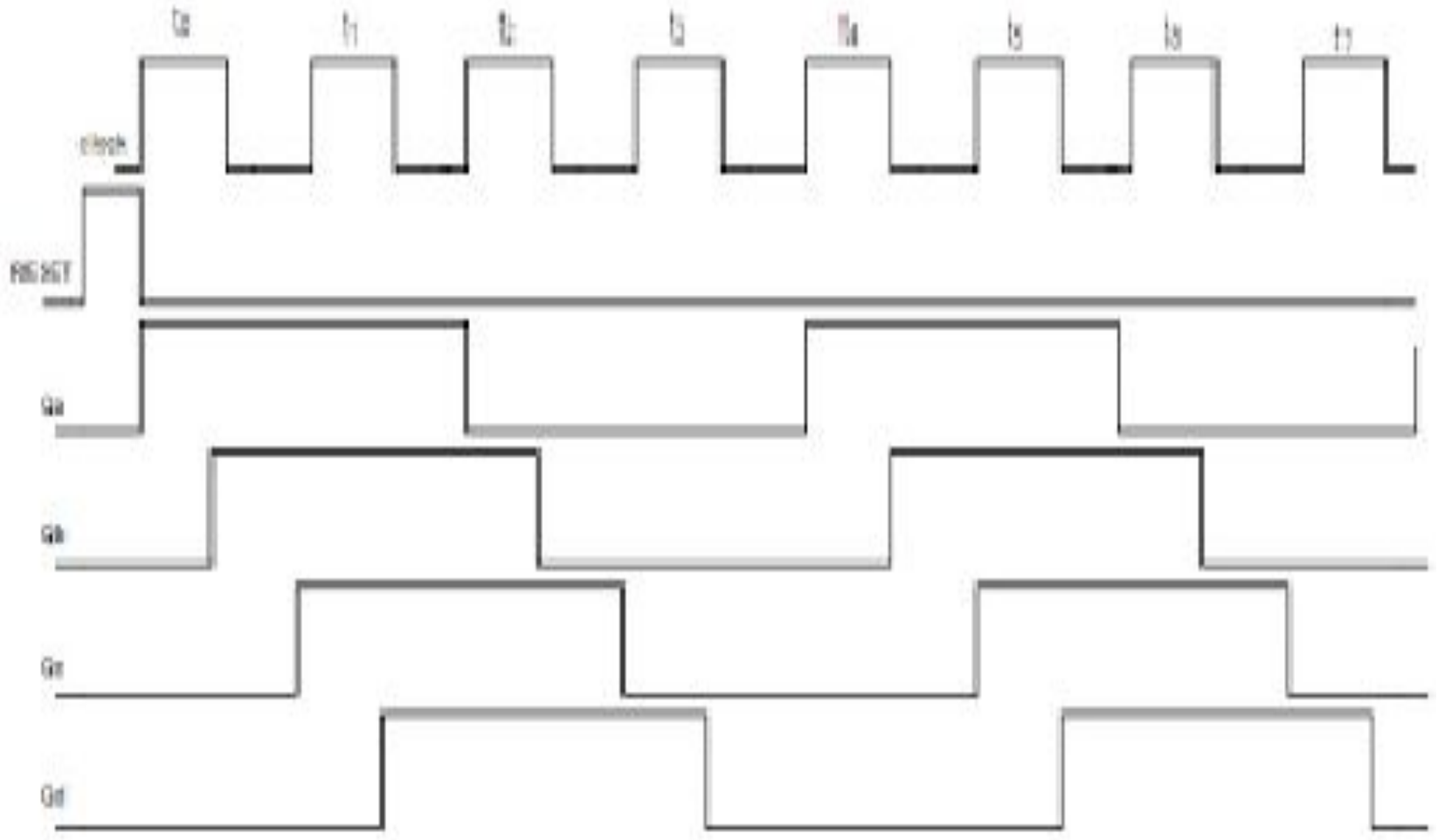
A Johnson counter is basically a shift register counter in which the output of the first flip flop is connected to the next flip flop and so on and the inverted output of the last flip flop is again fed back to the input of the first flip flop. They are also known as twisted ring counters. The logic circuit given below shows a Johnson Counter. The circuit consists of four D flip-flops which are connected. An n-stage Johnson counter yields a count sequence of 2n different states, thus also known as a mod-2n counter. Since the circuit consists of four flip flops the data pattern will repeat every eight clock pulses as shown in the truth table below:

| Clock Pulse No | FFA | FFB | FFC | FFD |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 |
| 2 | 1 | 1 | 0 | 0 |
| 3 | 1 | 1 | 1 | 0 |
| 4 | 1 | 1 | 1 | 1 |
| 5 | 0 | 1 | 1 | 1 |
| 6 | 0 | 0 | 1 | 1 |
| 7 | 0 | 0 | 0 | 1 |




The main advantage of Johnson's counter is that it only needs n number of flip-flops compared to the ring counter to circulate a given data to generate a sequence of 2n states.

# JOHNS RING COUNTER TIMING DIAGRAM



# SEQUENCE DETECTOR

A **sequence detector** is a **sequential** state machine which takes an input string of bits and generates an output 1 whenever the target **sequence** has been detected.

## Overlapping
## Non-Overlapping

In an overlapping sequence detector the last bit of one sequence becomes the first bit of next sequence. However, in non-overlapping sequence detector the last bit of one sequence does not become the first bit of next sequence

For non overlapping case
Input :0110101011001
Output:0000100010000
For overlapping case
Input :0110101011001
Output:0000101010000

## Mealy machine

In a Mealy machine, output depends on the present state and the external input (x). Hence in the diagram, the output is written outside the states, along with inputs

## Moore machine

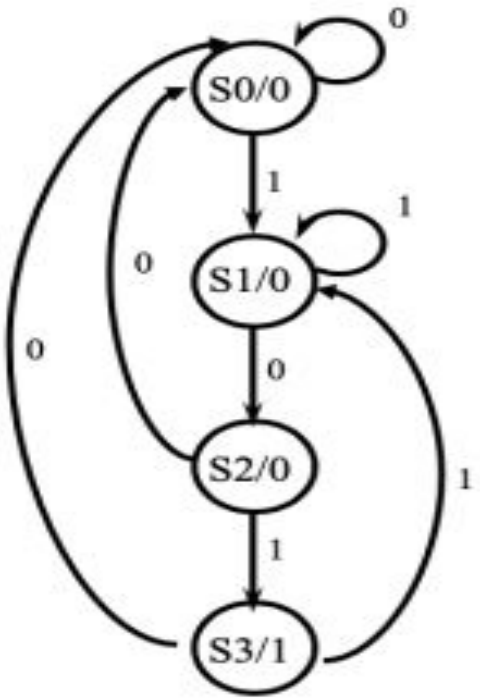
In a Moore machine, output depends only on the present state and not dependent on the input (x). Hence in the diagram, the output is written with the states.

Develop Mealy machine for a 101 sequence detector.

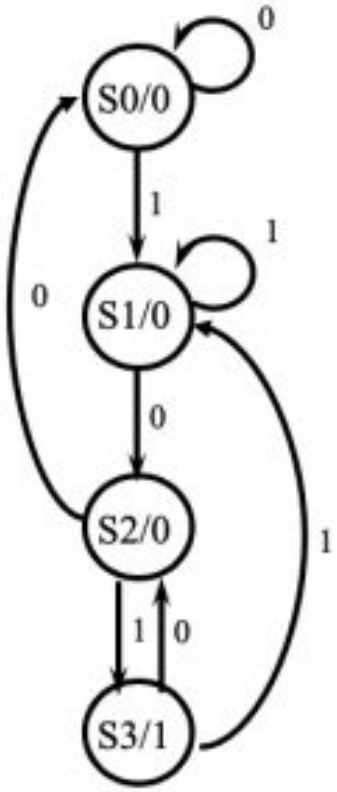
# *SEQUENCE DETECTOR 101*

# MOORE MACHINE

## Non-Overlapping

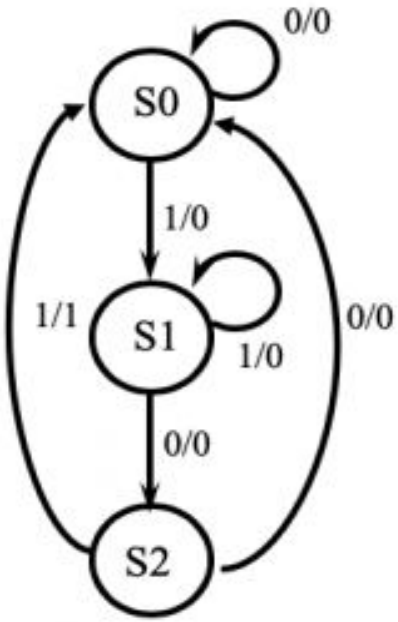



## Overlapping

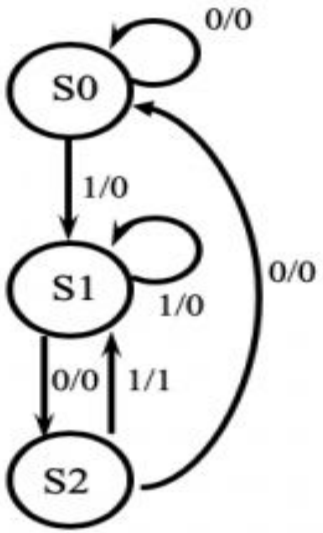


# MEALY MACHINE

## Non-Overlapping




## Overlapping



**1.Develop the state diagram**

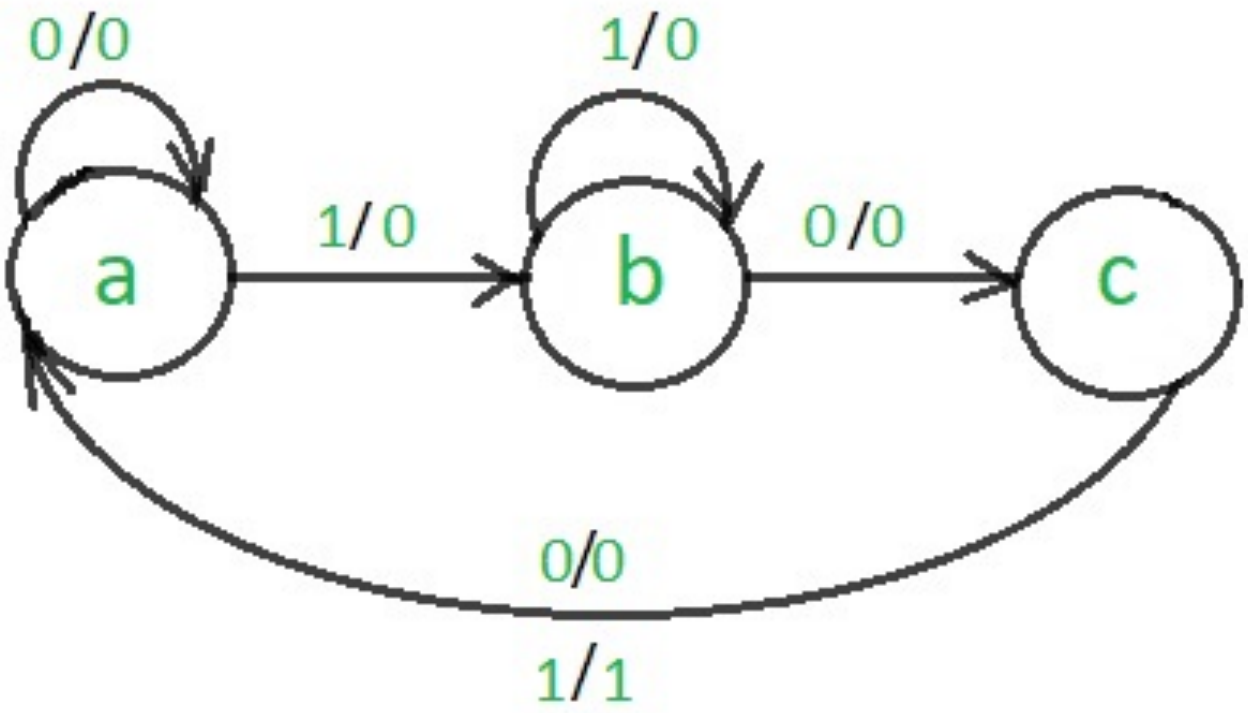
**2. Code Assignment –** States having the same next states for a given input condition should have adjacent assignments.

**3. Make Present State/Next State table**

**4. Draw K-maps for Dx, Dy and output (Z)**

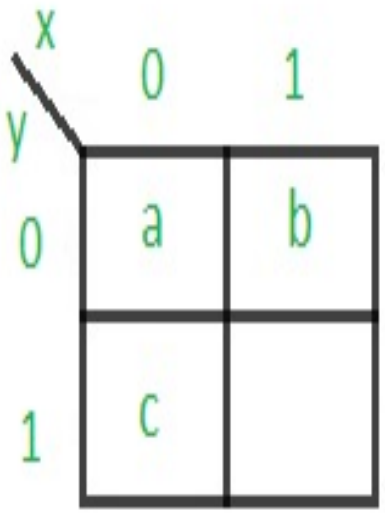
**5. Finally implement the circuit**

# STATE DIAGRAM



# CODE ASSIGNMENT

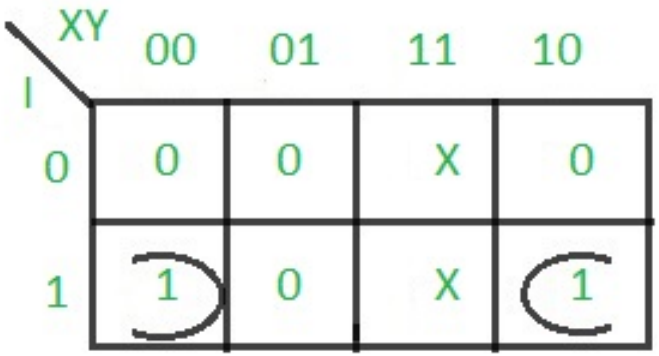
| Previous States | States | Next States |
|---|---|---|
| a,c | a | a,b |
| b,a | b | b,c |
| b | c | a |



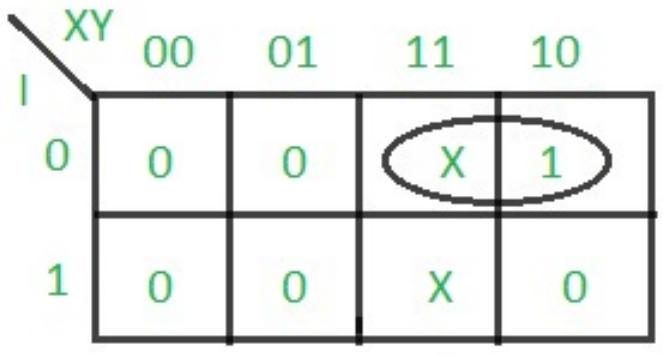
# PRESENT STATE/NEXT STATE TABLE

| Present States | | i/p | Next States | | Flip Flop Excitations | | O/P |
|---|---|---|---|---|---|---|---|
| X | Y | | X' | Y' | Dx | Dy | |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 | 1 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |
| 1 | 0 | 1 | 1 | 0 | 1 | 0 | 0 |
| 1 | 1 | 0 | X | X | X | X | X |
| 1 | 1 | 1 | X | X | X | X | X |

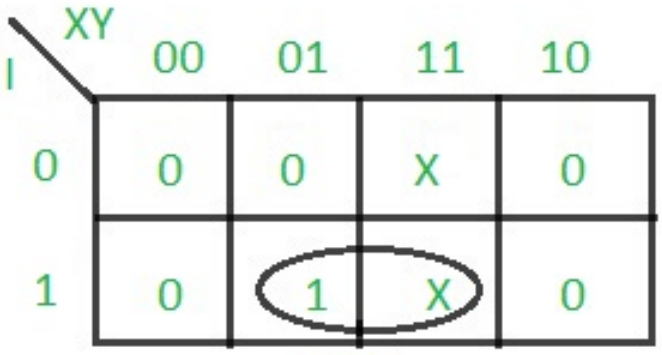
# DRAW K-MAPS




$Dx=\overline{Y}.I$




$Dy= X.\overline{I}$




$Z=Y.I$

# IMPLEMENT THE CIRCUIT

